

Four Wheel Drive (4WD) Systems



Electronic Shift-On-The-Fly (ESOF) System

The Electronic Shift-On-The-Fly (ESOF) system consists of the following components:

- Mode Select Switch (MSS) , located on the instrument panel
- Transfer Case Control Module (TCCM) , located near the passenger kick panel
- Integrated Wheel End (IWE) solenoid, located in left rear of the engine compartment
- Integrated Wheel Ends (IWEs) , located at each front wheel hub
- Transfer case assembly (includes shift motor and synchronization clutch)

Vehicles equipped with an ESOF system allow the operator to choose between 2-Wheel Drive (2WD) (2H) and 2 different Four-Wheel Drive (4WD) modes. The transfer case is shifted electronically based on the MSS position. Under normal driving conditions, the transfer case is in 2WD (2H), but when desired, the operator may shift into 4WD high (4H) or 4WD low (4L). The operator can shift between 2WD (2H) and 4WD high (4H) at any speed. When shifting into or out of 4WD low (4L) range, the TCCM requires that the vehicle speed is less than 5 km/h (3 mph) and the transmission in NEUTRAL.

The transfer case is equipped with an electronically controlled clutch which is located inside the case. This clutch is used to synchronize the speed of the front driveline with the rear driveline during 2WD (2H) to 4WD high (4H) shifts.

2-Speed Torque-On-Demand

The 2-speed torque-on-demand system consists of the following components:

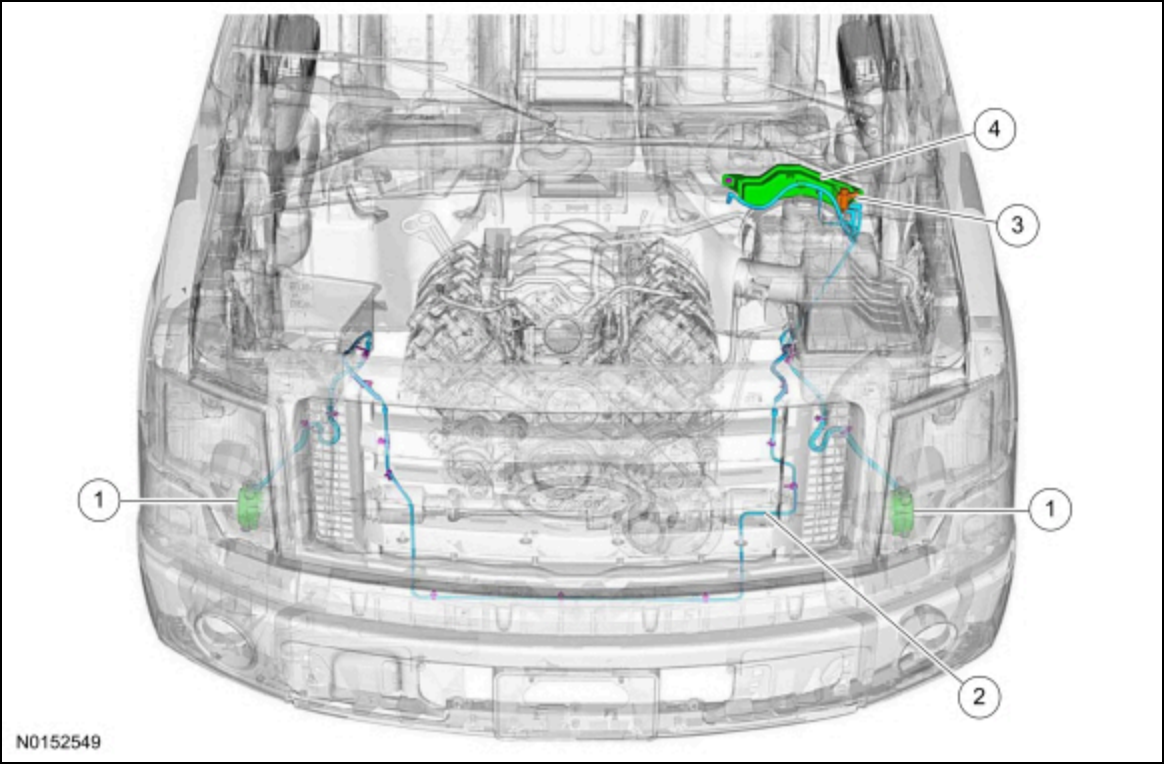
- MSS , located on the instrument panel
- TCCM , located near the passenger kick panel
- IWE solenoid, located in left rear of the engine compartment
- IWEs , located at each front wheel hub
- Transfer case assembly (includes shift motor and synchronization clutch)

Vehicles equipped with the 2-speed torque-on-demand system allow the operator to choose between 2WD and 3 different 4WD modes. The transfer case is shifted electronically based on the MSS position. Under normal driving conditions, the unit is in 2WD (2H), but when desired, the operator may shift into 4WD AUTO (4A), 4WD high (4H), or 4WD low (4L). When shifting into or out of 4WD (4L) range, the TCCM requires that the vehicle speed is less than 5 km/h (3 mph) and the transmission is in NEUTRAL.

The transfer case is equipped with an electronically controlled clutch which is located inside the case. This clutch is used to synchronize the speed of the front driveline with the rear driveline during 2WD (2H) to 4WD high (4H) or 4WD AUTO (4A) shifts. The clutch also provides torque to the front driveline in all 4WD modes.

Integrated Wheel End (IWE) System

The IWE system consists of the following components:



1	IWEs (spring-loaded vacuum hubs)
2	Vacuum hoses
3	IWE solenoid
4	Vacuum reservoir

The IWE system, used on ESQF and 2-speed torque-on-demand equipped vehicles, uses vacuum hubs that engage or disengage the front wheel hubs from the front halfshafts.

The IWE solenoid receives engine vacuum from the vacuum reservoir. When the 4WD system is in 2H mode, the TCCM supplies a ground path to the IWE solenoid to apply vacuum to the IWEs (disengaging the front hubs from the front halfshafts). When operating in any 4x4 mode, the TCCM does not supply the ground path to the IWE solenoid, vacuum is not applied to the IWEs and an internal spring keeps the front hubs engaged to the front halfshafts. Vehicles equipped with a 3.5L GTDI engine are equipped with an engine mounted/cam-driven brake vacuum pump that supplies vacuum to the IWE vacuum reservoir. The brake vacuum pump is located on the backside of the RH cylinder head. Refer to Section 206-07. As a first step in service, eliminate obvious items such as loose wiring connections, loose vacuum connections or damaged vacuum lines.

Electronic Locking Differential (ELD) System

The Electronic Locking Differential (ELD) system consists of the following components:

- ELD field coil, located at the rear differential
- Rear differential
- MSS (unique for ELD equipped vehicles, the ELD switch is integrated with the MSS), located on instrument panel

The optional ELD system, allows the operator to lock the rear differential side gears on slippery and/or off road surfaces. ELD operation is requested by pulling out on the MSS on 4X4 equipped vehicles, and by rotating the switch from OFF to ON on 4x2 vehicles.

Off-Road Mode

NOTE: Off-road mode is specifically calibrated for off-road driving conditions and should never be used on pavement.

The SVT Raptor comes equipped with a special off-road function. An off-road button is located on the floor console. A light on the button illuminates when this feature is activated.

The off-road function consists of the following components:

- Off-road mode switch
- ~~TCCM~~

Off-road mode affects 4 vehicle functions:

- It allows unique transmission shifting that optimizes off-road performance.
- Off-road mode provides optimized throttle response tailored for off-road performance driving.
- When off-road mode is engaged, the standard vehicle speed thresholds for the ~~ELD~~ are lifted.
- When off-road mode is engaged, the AdvanceTrac settings are altered for optimized off-road performance.

Neutral Flat Tow

NOTICE: ~~2WD~~ vehicles cannot be towed with any wheels on the ground as vehicle or transmission damage may occur.

NOTICE: Failing to put the transfer case in the NEUTRAL position will damage vehicle components.

Vehicles equipped with a Four-Wheel Drive (4WD) system have neutral flat tow software in the ~~TCCM~~. Following the neutral flat tow activation procedure commands the transfer case shift motor into a neutral position. Locking the transfer case in neutral will prevent damage to the transmission while towing a ~~4WD~~ vehicle on all 4 wheels (such as when being towed behind a motorhome).